Project Design Phase

Introduction

The Project Design Phase focuses on defining how the system will be structured and how different components of the Automated Ticket Assignment System will interact with each other. In this phase, the system architecture, database structure, user roles, and workflow automation are planned in detail before implementation.

The design phase ensures that the system can effectively automate ticket assignment while maintaining security, efficiency, and scalability. Using the ServiceNow platform, this project is designed to manage support tickets, store issue information, and automatically route tickets to the appropriate support group.

By carefully designing the system components, the project ensures smooth interaction between users, groups, roles, database tables, and automated workflows.

System Architecture

The Automated Ticket Assignment System follows a structured architecture where different components work together to manage support operations.

The architecture includes the following elements:

• **User Interface Layer** – Allows users to create and view support tickets through forms in the ServiceNow platform.

• **Application Layer** – Handles the business logic of the system, including ticket processing and automated assignment rules.

• **Database Layer** – Stores all ticket-related data such as issue details, assigned groups, and ticket status.

• **Automation Layer** – Uses Flow Designer to automatically assign tickets to the appropriate support group based on issue type.

This architecture ensures that tickets move efficiently from creation to resolution without manual intervention.

Database Design

The database design defines how information related to support tickets will be stored and managed within the system.

A custom table called **Operations Related** is created to store ticket information. This table acts as the central data repository for all support requests.

The main fields in the table include:

• Issue
• Description
• Assigned To
• Status
• Ticket Raised Date
• Priority

The **Issue field** contains predefined choices such as:

• Unable to login to platform
• 404 error
• Regarding certificates
• Regarding user expired

These predefined options help standardize the ticket creation process and enable automated routing of tickets to the correct support groups.

Role-Based Access Design

Role-based access control is implemented to ensure that only authorized users can access and manage ticket data.

The main roles used in the system include:

Platform Role
Responsible for handling issues related to platform errors such as login failures or system errors.

Certificate Role
Responsible for handling issues related to certificate generation or verification.

These roles are assigned to specific groups and users so that only relevant support members can access the tickets assigned to them.

Group Design

Groups represent support teams responsible for handling specific types of issues.

Two main groups are created in this project:

Platform Support Group
Handles issues related to platform functionality such as login errors, system errors, or user account problems.

Certificate Support Group
Handles issues related to certification processes and certificate-related queries.

When tickets are created, automation rules ensure that the ticket is assigned to the correct group based on the issue type.

Workflow Design

The workflow design focuses on automating the ticket assignment process using ServiceNow Flow Designer.

Two main automated flows are implemented in this project.

Certificate Ticket Assignment Flow

When a ticket is created with the issue type **"Regarding Certificates"**, the system automatically assigns the ticket to the Certificate Support Group.

Platform Ticket Assignment Flow

When a ticket is created with issues such as:

• Unable to login to platform
• 404 error
• Regarding user expired

the system automatically assigns the ticket to the Platform Support Group.

These workflows eliminate the need for manual ticket routing and significantly improve the efficiency of support operations.

Security Design

Security is implemented using **Access Control Rules (ACLs)**.

ACLs restrict access to sensitive data fields in the ticket table. Only users with the required roles can read or modify ticket information. This ensures that the system maintains data integrity and protects important information.